CFA Level 3 — Asset Allocation: Complete Breakdown

Source: Mark Meldrum 2026 Level 3 Notes (Last Revised 04/23/2025)

This document covers 5 learning modules plus integrated notes from your mock exam mistakes where they relate to asset allocation topics.

MODULE 1: Capital Market Expectations — Part 1: Framework & Macro Considerations

LOS a — The CME Framework (7 Steps)

Capital market expectations are projections about asset classes that feed directly into asset allocation decisions. The framework is a structured 7-step process:

1. **Specify the expectations needed** — define the asset class universe (geography, class, sub-class) and the time horizon (e.g., "U.S. equities over the next 5 years").
2. **Research the historical record** — identify factors that have driven returns historically and establish possible ranges.
3. **Specify methods/models** — e.g., CAPM, Fama-French. The ERP should reflect the investor's time horizon.
4. **Determine best data sources** — evaluate calculation methodologies, data collection techniques, error rates, potential biases.
5. **Interpret the current investment environment** — this step requires experience and judgment.
6. **Produce the set of expectations** from Step 1.
7. **Monitor actual outcomes** and compare with expectations.

Key principle: Levels of returns matter more than precision. Expectations must be internally consistent (e.g., rAI > rE > rB) and intertemporally consistent for the specified horizon. Inconsistency leads to poor risk-return characteristics in the resulting asset allocation.

LOS b — Challenges in Developing CME Forecasts

This is a dense section covering 9 distinct challenges. High-quality forecasts are consistent, unbiased, objective, well supported, and have minimum forecast error.

1. Limitations of economic data: Definition changes, construction methodology, timeliness, accuracy, revisions, re-basing, and time lags all compromise data reliability.

2. Data measurement errors and biases:

- *Transcription errors* — simple collection/entry mistakes (e.g., 3.1% vs 1.3%)
- *Survivorship bias* — presents an overly optimistic picture of returns

- *Appraisal (smoothed) data* — for assets without liquid public markets. This is critical: smoothed data produces lower correlations with other assets AND lower standard deviations, therefore **underestimates risk**. This comes up repeatedly in the timber/real estate questions.

> 🔴 **YOUR MOCK MISTAKE (Mock 8 — Timber):** You noted that *"both volatility and correlation measures will promote an upward bias (overstatement) of the risk-reduction benefits"* for timberland. This is exactly the appraisal/smoothed data problem — because private real assets trade infrequently, their reported volatility and correlations are artificially low, making them *appear* to offer more diversification benefit than they actually do. When the exam asks about the implications of including timberland or private real estate in an MVO, remember: the optimizer will over-allocate to these asset classes because the inputs are biased downward.

3. Limitations of historical estimates: These are starting points only. Regime changes (technological, political, legal/regulatory) mean historical data may cover multiple regimes (non-stationarity). Long data series increase precision of population parameter estimates but may span fundamentally different environments.

4. Ex-post risk as biased measure of ex-ante risk: Historical risk may reflect fears that never materialized — returns show the effect, but economic data does not.

5. Biases in analyst methods: Data-mining bias, time-period bias. Remedy: perform out-of-sample tests.

6. Failure to account for conditioning information: Are there relevant new facts in the present that should adjust forward-looking forecasts?

7. Misinterpretation of correlation: Correlation may be spurious, does not imply causation, and even if causal, doesn't indicate direction. Also: zero linear correlation doesn't preclude a significant non-linear relationship.

8. Psychological traps:

- *Anchoring* — initial impressions anchor subsequent judgments
- *Status quo* — perpetuate recent observations, predict no change
- *Confirming evidence* — seek out/overweight confirming evidence
- *Overconfidence* — predict too narrow a range of possibilities
- *Prudence* — temper forecasts to avoid appearing extreme
- *Availability bias* — overly influenced by events that left a strong impression

9. Model uncertainty: Whether the model is structurally/conceptually correct. Includes parameter uncertainty (estimated with error) and input uncertainty (are the inputs correct?).

LOS c — Exogenous Shocks

Events from outside the economic system that change its course. They are typically not built into prices (unanticipated or only partially anticipated). Can be positive or negative for growth.

Six categories: (1) policy changes, (2) new products/technology, (3) geopolitics, (4) natural disasters — typically short-term; old capacity replaced with more efficient facilities, (5) natural resources/critical inputs, (6) financial crises (typically considered endogenous but treated here).

LOS d — Economic Growth Trend Analysis

Trend growth is the average growth rate around which the economy cycles. Growth decomposes into:

1. **Growth from labour inputs** — growth in potential labour force size + growth in actual labour force participation
2. **Growth from labour productivity** — growth from increasing capital inputs + growth in total factor productivity (TFP)

Extrapolating past trends provides a reasonable initial estimate. The average level of default-free yields is pulled towards the trend growth level over sufficiently long horizons.

Equity valuation link: V(equity) = GDP × S(share of profit in economy, ~6%) × PE. Expected equity return = capital appreciation + dividend yield.

> 🔴 **YOUR MOCK MISTAKE (Mock 2 — GDP Forecast):** You got tripped up on "Calculate the projected long-run equity return for Country B." The long-run equity return from the GGM simplifies to: E(Re) = D/P + g, where g = nominal GDP growth (= real GDP growth + inflation). Make sure you're using the correct growth decomposition and not mixing real/nominal.

LOS e — Approaches to Economic Forecasting

Three distinct approaches:

1. **Econometric modelling** — quantitative, grounded in economic theory. Structural or reduced-form models. Exogenous variables supplied by the forecaster.
2. **Economic indicators** — leading, coincident, lagging. Usually combined into a diffusion index. Caveat: revisions make the index appear better at predicting than it really is. New method: *nowcasting* (e.g., GDPnow for current-quarter GDP).
3. **Checklist approach** — forecaster asks a series of questions. Involves substantial subjective judgment.

LOS f — Business Cycle Phases & CME Effects

Five phases with distinct implications for each asset class:

Phase	Economy	Short Rates	Long Rates	Stocks	Yield Curve
Initial Recovery	Picks up from recession; confidence low	Bottoming	Bottoming	Rally sharply; cyclicals/HY/EM outperform	Steep (steepest just before)
Early Upswing	Confidence grows; unemployment falls	Begin rising	Stable	Rising	Front steepening, back flattening
Late Upswing	Output gap closed; inflation looming	Rising	Rising	Rising + volatility increases	Flattening from long inwards
Slowdown	Economy slowing; inventory downswing	Topping out	Bonds rally	Soft/falling	Flat to inverted
Recession	Inventory pullback; investment drops	Dropping	All yields drop	Drop; bottom before end of recession	Steepening

LOS g — Inflation and Asset Class Returns

Inflation is procyclical, as are inflation expectations.

- **Cash:** Zero duration, floating-rate, inflation-protected. Attractive in rising rate environments.
- **Bonds:** Rising inflation = higher discount rates = lower prices. Deflation benefits IG bonds, hurts HY.
- **Stocks:** At expectation → little effect. High → higher rates slow economy. Low → decline in asset prices.
- **Real Estate:** Rents and property values rise with inflation. Low inflation → declining asset prices.

> 🔴 **YOUR MOCK MISTAKE (Mock 5 — Interest Rates & Inflation):** "Which of the following is most likely to occur if the portfolio manager's concerns about monetary and fiscal policies for Country B are realized?" Remember: if policy is expansionary beyond what's sustainable, expect higher inflation → bonds negative, real estate positive (rents rise), equities mixed depending on pricing power.

LOS h — Monetary & Fiscal Policy

Central banks moderate cyclical behaviour. Easier to slow the economy than stimulate it.

Taylor Rule: $r(\text{target}) = r(\text{neutral}) + \pi(e) + 0.5(\text{GDP}(e) - \text{GDP}(\text{trend})) + 0.5(\pi(e) - \pi(\text{target}))$

All four variables (r(neutral), π(e), GDP(trend), π(target)) are **unobservable** — this is a critical exam point.

Long-run CME should use r(neutral). Short-term: consider the expected path of rates, but historical data become less reliable and the yield curve shape may be unreliable (engineered through QE/negative rates).

> 🔴 **YOUR MOCK MISTAKE (Mock 2 — Taylor Rule):** You missed this calculation. The Taylor Rule is a nominal rate — make sure you're adding expected inflation as a separate component alongside the neutral rate. If the output gap is negative (GDP below trend), the Taylor Rule output will be below the neutral rate, implying policy should be accommodative.

LOS i — Yield Curve as Economic Predictor

The shape of the yield curve changes systematically through the business cycle:

- Initial Recovery → **steep**
- Early Expansion → front steepening, back-half flattening
- Late Expansion → **flattening** from long end inwards
- Slowdown → **flat to inverted**
- Contraction → **steepening**

> 🔴 **YOUR MOCK MISTAKE (Mock 5 — Yield Curve):** "The yield curve for Country B's default-free government bonds is most likely..." You noted "IG bonds are always upward sloping." Be careful here — the question is about **government** bonds. The yield curve for default-free bonds follows the business cycle pattern above. An inverted curve signals slowdown/recession expectations. IG corporate curves are typically upward sloping because of the credit premium, which is a different concept.

LOS j — International Interactions

Macroeconomic linkages: (X − M) = (S − I) + (T − G). As the current account balance changes, interest rates, exchange rates, and asset prices adjust to keep the capital account in balance.

Interest rate/exchange rate linkages (impossible trinity): A country cannot simultaneously (1) allow unrestricted capital flows, (2) maintain a fixed exchange rate, and (3) pursue independent monetary policy. Two countries can share a yield curve only if capital flows are unrestricted AND the FX rate is credibly fixed *forever*.

MODULE 2: Capital Market Expectations — Part 2: Forecasting Asset Class Returns

LOS a — Fixed Income Return Forecasting

Discounted Cash Flow / YTM approach: If investment horizon < MacDuration → capital gains/losses dominate. If horizon = MacDuration → E(RB) ≈ YTM. If horizon > MacDuration →

reinvestment dominates.

Building Block Approach: One-period default-free rate + Term premium + Credit premium + Liquidity premium

Term premium drivers: (1) Level-dependent inflation uncertainty, (2) Ability to hedge recession risk — good hedge = low term premium (when AD drives growth & inflation, bond returns negatively correlated with growth), (3) Supply/demand at various maturities, (4) Cyclical effects — countercyclical with growth.

Credit premium: IG reflects credit migration risk. HY reflects default risk and is countercyclical. Credit spreads are driven primarily by credit premium + financial market conditions, only secondarily by expected default losses.

Liquidity premium: Lower for bonds priced near par, relatively new, from large/well-known issuers, with standard/simple structures, high quality.

LOS b — EM Fixed Income Risks

Economic risks (ability to pay): Concentration of wealth, dependence on cyclical industries, restrictions on trade/capital/currency, poor fiscal/monetary discipline, reliance on foreign borrowing.

Quantitative thresholds: Fiscal deficit/GDP > 4%, Debt/GDP > 70–80%, rGDP growth < 4%, CA deficits > 4% of GDP, foreign debt > 50% of GDP, forex reserves < 100% of short-term debt.

Political & legal risks (willingness to pay): Weak property rights, weak contract enforcement, sovereign immunity, confiscation risk, coalition governments.

> 🔴 **YOUR MOCK MISTAKE (Mock 3 — Currency Flows):** You were asked about "two additional risks posed by the international money flows to Country Y." You noted these are "hot money" / carry trade flows. Remember: hot money flows into EM economies are volatile capital inflows driven by yield differentials, not fundamental investment. Risks include: (1) sudden reversal/capital flight when spreads narrow or risk appetite falls, (2) currency appreciation that hurts export competitiveness, (3) asset price bubbles from excess liquidity.

LOS c — Equity Return Forecasting

1. Historical Statistics Approach: The variability in P/E and E/GDP ratios means sample averages are very imprecise. Consider shrinkage estimators.

2. Discounted Cash Flow / Grinold-Kroner Model:

$E(Re) = (D/P - \%\Delta S) + \%\Delta E + \%\Delta(P/E)$

Where D/P = dividend yield, %ΔS = share repurchases/issuance, %ΔE = earnings growth, %Δ(P/E) = repricing.

Long-term simplification: %ΔE = nominal GDP growth, %ΔS ≈ 0, %Δ(P/E) ≈ 0 (mean reverts). Therefore: **E(Re) = D/P + nominal GDP growth**.

> 🔴 **YOUR MOCK MISTAKE (Mock 8 — Equity Premium/EPS-GDP):** You noted: *"Change in EPS/GDP component already captures two things: (1) the change in the share of corporate*

> *profits relative to the total economy, (2) the impact of share buybacks (which increase EPS even if total profit stays the same)."* This is an important distinction. In the Grinold-Kroner decomposition, the share buyback effect is captured in %ΔS (negative %ΔS = buybacks = positive for return). Don't double-count it in the earnings growth component.

3. Risk Premium / Singer-Terhaar Approach:

Fully integrated: RP(i,G) = ρ(i,GM) × σ(i) × SR(GM)

Fully segmented: RP(i) = σ(i) × SR(i) — correlation term = 1 because the asset IS the market.

Blended: RP(i) = ω × RP(integrated) + (1−ω) × RP(segmented)

Where ω = degree of integration. Developed markets: 0.75–0.90. Emerging markets: 0.50–0.75.

LOS e — Real Estate Return Forecasting

Properties trade infrequently → must rely on appraisal data → return series is too smooth → **understates volatility and correlations** (does not affect the mean).

Cap Rate: r = NOI / Property Value

Long run: E(r) = NOI/P + g (where g ≈ GDP growth)

Short run: E(r) = NOI/P + g − %Δ(Cap Rate) — if cap rates rise, returns fall.

Cap rate drivers: positively related to interest rates and credit spreads (these partially offset each other); negatively related to credit availability.

Risk premium perspective: Default-free rate + term premium (long-lived asset) + credit premium (tenant risk) + equity premium (ownership risk) + **liquidity premium (2–4% for commercial RE)**.

REITs act like stocks in the short term but like real estate in the long run.

LOS f — Forecasting Exchange Rates

Three channels through goods & services:

1. *Trade flows* — do not exert significant impact (net flows typically small)
2. *PPP* — %ΔS ≈ π(f) − π(d). Poor predictor in short-to-intermediate term.
3. *Competitiveness/sustainability of current account* — CA deficit > 2% of GDP raises concerns unless driven by high investment.

Capital flows equation: E(%ΔS(d/f)) ≈ (r(d) − r(f)) + (Term(d) − Term(f)) + (Credit(d) − Credit(f)) + (Equity(d) − Equity(f)) + (Liq(d) − Liq(f))

If domestic country offers higher risk-adjusted returns across any of these segments, currency should appreciate.

UIRP does not hold in practice → carry trades are profitable (borrow low-yield, invest high-yield).

> 🔴 **YOUR MOCK MISTAKE (Mock 4 — FX Time Horizon):** "The most appropriate time horizon for Singh's exchange rate forecasting approach is..." Different approaches apply at different horizons: PPP works long-term only, capital flow models work short-to-intermediate, current account sustainability is medium-to-long term. Match the approach described in the vignette to the correct horizon.

LOS g — Forecasting Volatility

Five approaches:

1. **Sample statistics** — constant VCV matrix from historical data. Subject to substantial sampling error.
2. **VCV from multi-factor models** — handles many assets, less estimation error, improves cross-sectional consistency. But: models rarely correctly specified → biased and inconsistent (won't converge to true VCV).
3. **Shrinkage estimation** — weighted combination of sample VCV and target VCV (e.g., factor-based). Better estimate than either alone.
4. **De-smoothing** — for private assets. $R(t) = (1-\lambda)r(t) + \lambda R(t-1)$. True variance = $[(1+\lambda)/(1-\lambda)] \times$ var(observed) > var(observed).
5. **ARCH models** — time-varying volatility. Higher $(\alpha+\beta)$ → more volatility clustering.

LOS h — Portfolio Reallocation Based on Macro Factors

- ↑ **Trend growth** → favours equities, higher real rates → negative for bonds
- ↑ **Global integration** → required returns decline. Shift from developed to emerging.
- **Business cycle positioning** → approaching trough: buy equities, reduce bonds (steep curve = high term + credit premium). Barbell structure anticipating flattening.
- **Current accounts** → reallocate from countries with rising deficits to rising surpluses.
- **Capital accounts** → countries with undervalued assets may have overvalued currency, potentially negating the undervaluation.

MODULE 3: Overview of Asset Allocation

LOS a — Investment Governance

The structure ensuring assets are invested to achieve objectives within risk tolerances/constraints. Three levels: (1) Investment committee (board members), (2) Investment staff, (3) Third-party resources.

Six elements of effective governance:

1. Articulate long- and short-term objectives (return objective + risk tolerance)

2. Allocate decision rights/responsibilities effectively
3. Specify the process for developing/approving the IPS
4. Specify the process for developing/approving the SAA (investment committee typically retains approval)
5. Establish a reporting framework
6. Periodically undertake a governance audit (independent third party)

> 🔴 **YOUR MOCK MISTAKE (Mock 5 — Effective Investment Governance):** "Determine one practice of DBF that is not consistent with effective investment governance." You noted: *"A fundamental rule is that the people who execute a process should not be the same people who audit/verify that process."* This is the governance audit principle — Element 6 requires independent review. If the same committee that sets the SAA also conducts the governance audit, that's a violation.

LOS b — Economic Balance Sheet

Conventional assets and liabilities PLUS extended portfolio items: PV(future earnings/human capital), PV(pension income), PV(expected inheritances), PV(future consumption as a liability).

Human capital is approximately **30% equity-like and 70% bond-like** (with significant variation among industries). Younger investors have proportionally more human capital → can tolerate more equity in their financial portfolio. As human capital depletes with age, the financial portfolio should shift toward bonds.

LOS c — Three Approaches to Asset Allocation

Approach	Objective	Risk Measure	Best For
Asset-Only (MVO)	Max Sharpe ratio for acceptable σ	σ(P), tracking risk, VaR	Investors without explicit liabilities
Liability-Relative (LDI)	Fund liabilities; invest excess for growth	Shortfall risk, vol of contributions	DB pensions, insurance
Goals-Based	Achieve goals with specified probabilities	Probability of failing to achieve goal	Individuals & families

Key distinction: Institutional liabilities are legal obligations, uniform, numerous (law of large numbers applies). Individual goals are not legal obligations, are many and varied, and don't benefit from averaging.

LOS e-f — Asset Class Specification & Risk Factors

Greer's three super-classes: (a) Capital assets — ongoing source of value, (b) Consumable/transformable assets — commodities, (c) Store of value — currencies, gold, art.

Five criteria for asset class specification:

1. Assets within a class should be relatively homogeneous
2. Asset classes should be mutually exclusive
3. Asset classes should be diversifying (low ρ)
4. Should make up a preponderance of world investable wealth
5. Should have capacity to absorb a meaningful proportion of an investor's portfolio

> 🔴 **YOUR MOCK MISTAKE (Mock 1 — Asset Class Specification Criteria):** You flagged: *"Be prepared for general definitions."* The exam tests whether you can recite these 5 criteria and apply them. When a question asks whether a proposed asset class meets the criteria, check each one systematically — particularly mutual exclusivity (e.g., adding "global small-cap" when you already have "global equities" creates overlap) and diversification benefit.

Factor-based approaches: Don't use asset classes as the unit of analysis. Instead, specify risk factors (market, size, value, momentum, liquidity, duration, credit, volatility) and assign investments based on factor exposures. Each factor is a zero-investment long-short portfolio with market exposure removed → low inter-factor correlations. Asset class allocation can obscure overlapping risk factors.

LOS g — Strategic Asset Allocation (8 Steps)

1. Quantify objectives
2. Establish risk tolerance
3. Determine investment horizon
4. Determine other constraints/requirements
5. Determine suitable AA approach (AO / LR / GB)
6. Specify asset classes + CME
7. Develop range of AA choices
8. Test/simulate potential stressors

LOS h — Global Market Portfolio

The cap-weighted portfolio of all investable assets. It minimizes non-diversifiable risk, serves as a reference point, provides discipline against home country bias. Investor must justify deviations from GMP weights.

LOS i-j — Implementation Choices & Rebalancing

TAA (Tactical): Short-term deviations from SAA. Active management at the asset class level.

Dynamic AA: Longer-term deviations motivated by valuation signals or economic views.

Rebalancing approaches:

- Calendar rebalancing — periodic (monthly, quarterly, annual)
- Percent-range rebalancing — trigger points around target weights

Factors affecting rebalancing ranges: higher transaction costs → wider; more risk averse → tighter; less correlated assets → tighter; belief in momentum → wider; belief in mean reversion → tighter; illiquid investments → problem; taxes → wider and asymmetric.

MODULE 4: Principles of Asset Allocation

LOS a — Mean-Variance Optimization

Objective function: Maximize $U(m) = E(R) - 0.005\lambda\sigma^2(m)$

Where λ = risk aversion coefficient. $\lambda=0$ → risk neutral (maximize return). $\lambda=4$ → moderately risk averse.

MVO is a **single-period framework** → CME should be matched with the evaluation horizon.

Efficient frontier properties: Corner portfolios are points where an asset class enters or leaves the efficient mix. The tangency portfolio has the highest Sharpe ratio. The global minimum variance portfolio has the lowest achievable risk.

Two treatments of cash: (1) As a risky asset in the MVO, or (2) as the risk-free asset, producing a linear efficient frontier through the tangency portfolio (two-fund separation).

Including extended assets: Human capital modelled as part bond-like, part equity-like. Real estate included using de-smoothed indexes. Non-tradeable assets are forced into the optimizer at their share of total wealth.

Criticisms of MVO (Critical Exam Content)

1. **Outputs highly sensitive to small changes in inputs** — a 50 bps change in expected return for one asset class can dramatically change the entire allocation.
2. **Allocations tend to be highly concentrated** in a subset of available asset classes.
3. **Investors care about more than mean and variance** — skew, kurtosis, tail risk matter. Alternative optimizations: mean-semivariance, mean-CVaR, mean-variance-skewness-kurtosis.
4. **Sources of risk may not be diversified** even if asset class weights appear diversified → use factors.
5. **Not connected to liabilities or goals** → use liability-relative or goals-based approaches.
6. **Single period; ignores trading costs and taxes** → use simulation.

> 🔴 **YOUR MOCK MISTAKE (Mock 8 — MVO Portfolio Selection):** "Which strategic asset allocation considered by Sanchez would be most appropriate for Baker?" When selecting among portfolios on the efficient frontier, you need to match the portfolio's risk level to the investor's risk tolerance (λ). A higher λ investor selects a portfolio further left (lower σ) on the frontier. Don't just pick the highest Sharpe ratio portfolio unless the question specifically asks for it.

> 🔴 **YOUR MOCK MISTAKE (Mock 4 — Choosing a Portfolio):** "Which allocation should Heart recommend to Gabbro?" Same concept — match λ to the utility function. Calculate U for

> each candidate allocation and pick the highest. Make sure you're using $0.005\lambda\sigma^2$ (with σ in percentage terms), not $0.5\lambda\sigma^2$.

Remedies:

- **Reverse optimization** — start with assumed optimal weights (market-cap weights) + covariances + λ, solve for implied expected returns. Then use CAPM: E(Ri) = rf + β(ERP).
- **Black-Litterman** — allows investor's own distinctive views to modify reverse-optimization outputs. Two methods: (a) absolute return forecast for an asset class, (b) return differentials between asset classes.
- **Resampled MVO** — Monte Carlo simulation to generate many sets of market assumptions, produce an efficient frontier for each, average the allocations.
- **Adding constraints** — specify allocation ranges, upper limits, relative allocations between classes.

LOS d — Allocating to Less Liquid Asset Classes

Problem: lack of accurate indexes (complicates CME), no passive vehicles to track them.

Three solutions: (1) Exclude from optimization; carve out a fixed allocation, (2) Include but model specific risk characteristics of likely implementation vehicles (factor-based), (3) Include using listed proxies (listed RE, listed infrastructure, public equity proxies for PE) — may violate mutual exclusivity.

LOS e — Risk Budgeting

Identify total desirable portfolio risk (σ(P)), then allocate efficiently across asset classes.

MCTR (Marginal Contribution to Total Risk) = β(i) × σ(P) — the rate at which risk changes with a small change in weight.

ACTR (Absolute Contribution to Total Risk) = w(i) × MCTR(i). Sum of all ACTRs = σ(P).

Optimality condition: (E(Ri) – rf) / MCTR is the same for all asset classes AND matches the Sharpe ratio of the total portfolio.

> 🔴 **YOUR MOCK MISTAKE (Mock 4 — Risk Parity):** "Based on the risk parity approach, the weight that emerging market equities should have in the portfolio is..." In risk parity, each asset's ACTR must be equal: w(i) × MCTR(i) = (1/n) × σ^2(P). This means lower-volatility assets get higher weights. Don't confuse risk parity with equal weighting (1/N).

LOS h — Factor-Based Asset Allocation

Factors: market, size, valuation, momentum, liquidity, duration, credit, volatility. Each constructed as a zero-investment portfolio (long-short) with market exposure removed → low inter-factor correlations. To convert back to total return space, add rf.

LOS j-l — Liability-Relative Asset Allocation

Surplus = MV(assets) – PV(liabilities). Funding ratio = MVa / PVL. Both are highly dependent on the discount rate used.

Characterizing liabilities: (1) Fixed vs. contingent cash flows, (2) Legal vs. quasi-liabilities, (3) Duration & convexity, (4) Size relative to sponsoring organization, (5) Factors driving future cash flows, (6) Timing/longevity risk, (7) Regulations.

Three approaches:

1. Surplus Optimization (one-fund): Extension of MVO optimizing over surplus. Liabilities enter as a "negative asset." Can handle any funding status and any risk level. At low risk levels, the allocation resembles the H/RS portfolio.

2. Hedging/Return-Seeking Portfolio (two-fund): (a) Hedging portfolio — assets driven by the same factors as the liabilities (duration matching, CFM). (b) Return-seeking portfolio — optimized via MVO. Most appropriate for conservative investors or when PVL is large relative to the firm. If funding ratio < 1, cannot create a fully hedged portfolio without additional contributions.

3. Integrated Asset-Liability Approach: When liability composition decisions must be made alongside asset allocation (e.g., banks, insurance companies). Often implemented as multi-period models.

> 🔴 **YOUR MOCK MISTAKE (Mock 2 — Pension Plan Approach):** Make sure you can match the appropriate liability-relative approach to the investor's situation. Surplus optimization is most flexible, H/RS is for conservative plans with large liabilities, integrated is for entities that can change both sides of the balance sheet.

LOS m — Goals-Based Asset Allocation

Pre-established set of risk-return modules (A through E on an efficient frontier). Each goal has its own time horizon and probability of success.

Goal classification: Needs (90–99% PoS), Wants (80–90%), Desires (60–80%).

Discount rate / funding cost: The minimum return required to meet a goal with the specified probability over the specified horizon. More critical goals → higher PoS → lower discount rate → **greater capital required**.

> 🔴 **YOUR MOCK MISTAKE (Mock 3 — Goal Portfolios):** You noted: *"Lowest funding cost means highest expected return."* This is exactly right — the module with the highest expected return adjusted for the required PoS and time horizon will require the least capital (lowest funding cost) to defease the goal. Select the module that offers the highest probability-adjusted return for the goal's horizon.

> 🔴 **YOUR MOCK MISTAKE (Mock 7 — Goal-Based Portfolio):** You noted *"I used solved for Present Value when I should have gotten PMTs."* When the goal involves a series of future payments (annuity) rather than a lump sum, you need to discount the payment stream using the

> appropriate discount rate from the module, not calculate a simple PV. The funding cost is the required initial capital to fund the series of cash flows.

LOS n — Heuristic Approaches

- **120 minus age** = equity allocation. Results in linear decrease in equity exposure. Close to target-date fund glide paths.
- **60/40 stock/bond** — evidence that the global financial asset portfolio is close to this split.
- **Endowment model** — emphasizes large allocations to alternatives; seeks illiquidity premium.
- **Risk parity** — each asset contributes equally to total risk. Tends to ignore returns and allocate entirely on risk. Uses leverage to achieve desired σ(P).
- **1/N rule** — equal weight across N asset classes. Empirically performs considerably better than theory suggests.

LOS o — Rebalancing (Expanded)

Rebalancing earns a **diversification return**: the compound growth rate of the rebalanced portfolio exceeds the weighted average compound growth rates of the components. Economically, rebalancing is equivalent to being short volatility (sell calls at upper range, sell puts at lower range).

Factors favouring **narrower** ranges: low transaction costs, higher risk aversion, lower correlation with rest of portfolio, higher volatility of the rest of the portfolio, belief in mean reversion, liquid assets.

Factors favouring **wider** ranges: higher transaction costs, lower risk aversion, higher correlation, belief in momentum, illiquid assets, taxes (encourage wider AND asymmetric ranges).

Volatility-based rebalancing: Corridor width = x × σ(R(A)). Sets corridor proportional to asset volatility.

MODULE 5: Asset Allocation with Real-World Constraints

LOS a — Asset Size, Liquidity, Time Horizon, Regulatory

Asset size: Too small → can't meet minimums, insufficient governance capacity. Too large → may exhaust capacity of active managers.

Liquidity: Varies by goals and time horizon. Allocations to illiquid classes influenced by available financial resources.

Time horizon: Defined by future liabilities/goals. As time passes, allocations must reflect shortening horizons even if risk preferences don't change. Lower risk/return assets for short-term goals; higher risk/return for long-term goals.

> 🔴 **YOUR MOCK MISTAKE (Mock 3 — Constraints):** "Discuss two constraints the Yangs would most likely face regarding the fund investment." You identified: *"Liquidity — portfolio is*

> *too small — time horizon."* Three constraints to consider for individual investors with concentrated alternative allocations: (1) Liquidity — PE/HF lock-ups vs. spending needs, (2) Asset size — minimum investments may consume too large a share of the portfolio, preventing diversification, (3) Time horizon — illiquid investments require long commitments that may not match the investor's planning horizon.

LOS b — Tax Considerations

After-tax returns differ by asset type: interest taxed highest, dividends lower, capital gains lowest (in most jurisdictions).

After-tax portfolio optimization: Adjust each asset class's E(R) and σ for expected taxes.

- Interest: r(at) = r(pt) × (1 – t)
- Equity: r(at) = P(d) × r(pt) × (1 – t(d)) + P(a) × r(pt) × (1 – t(cg))
- After-tax σ = σ(pt) × (1 – t) — taxes mute volatility.

Critical insight: Since after-tax volatility < pre-tax volatility, it takes larger asset class movements to materially alter the risk profile → **rebalancing ranges for taxable portfolios can be wider**.

Tax-efficiency strategies: Tax-loss harvesting, tax location (less tax-efficient assets in tax-advantaged accounts, low-tax assets in taxable accounts).

> 🔴 **YOUR MOCK MISTAKE (Mock 6 — Tax Drag):** "Calculate the tax drag (as a % of investment gains) on Stenson's portfolio over the 8-year period." Tax drag = [return without taxes – return with taxes] / return without taxes. Make sure to compute the compounded after-tax return correctly over the full period and remember that for capital gains only, the tax drag can't exceed the tax rate (you noted: *"must remove the 1 at the end because it is capital gains only — can't be less than the tax rate"*).

LOS c — Revising the SAA

Triggered by changes in: (1) Goals — personal circumstances, business conditions, (2) Constraints — time horizon, liquidity, asset size, legal/regulatory, (3) Beliefs — CME changes, new committee members, (4) Passage of time — natural glide path adjustments.

> 🔴 **YOUR MOCK MISTAKE (Mock 2 — Equity Allocation Change):** "Based on the unexpected change in Rahim's financial situation, the percentage of equities in his overall portfolio should..." When an investor's circumstances change (e.g., sudden large inheritance, job loss, new liability), re-evaluate the economic balance sheet. A reduction in human capital or increase in near-term spending needs typically means reducing equity exposure. An increase in financial capital with no change in risk tolerance might increase equity exposure in absolute terms but not necessarily as a percentage.

LOS d — Tactical Asset Allocation

TAA = deliberate short-term deviations from SAA to exploit business cycle opportunities or temporary price dislocations. Done at the asset class level, an asset-only approach.

Two types: (1) Discretionary TAA — manager skill in predicting/timing, (2) Systematic TAA — technical analysis, market signals (value, momentum, trend-following).

Evaluate TAA success: Compare Sharpe ratio vs. policy portfolio, evaluate information ratio, plot realized return/risk vs. SAA efficient frontier. TAA incurs additional trading costs and taxes, and may increase concentration risk.

LOS e — Behavioral Biases in Asset Allocation

Bias	Effect on AA	Mitigation
Loss aversion	Interfere with maintaining SAA through drawdowns	Goals-based investing (high-priority goals in low-risk assets)
Illusion of control	Extreme TAA, excessive trading, leverage, eliminating asset classes	Use GMP as starting point
Mental accounting	Segregating wealth into separate buckets	Goals-based investing incorporates this constructively
Representative bias	Return chasing in TAA	Objective SAA process, strong governance framework
Framing bias	AA influenced by how risk/return is presented	Present risk in multiple ways
Availability bias	Home country bias	Use GMP as starting point

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MOCK MISTAKES — Additional Asset Allocation Notes

Beyond the topic-specific mistakes integrated above, here are your remaining AA-related mock errors:

Mock 6 — Fed Rate Probability: "Calculate the probability of a 25 bps rate hike at the next meeting." This ties to CME Part 1 — the implied probability from fed funds futures: P = (Implied Rate – Current Rate) / 0.25. Make sure you're reading the futures price correctly and converting to an implied rate.

Mock 5 — Important Portfolio Selection: You flagged this as a significant error but didn't add detail. The general principle: when selecting a portfolio for a specific investor, always check (1) does it satisfy all constraints, (2) does the risk level match λ, (3) is the Sharpe ratio competitive with alternatives at the same risk level. Don't just optimize on return — the utility function penalizes risk.

Mock 2 — Initial Capital (Goals-Based): The initial capital required = PV of future goal cash flows, discounted at the rate associated with the chosen module's probability-adjusted return for the given time horizon. Higher PoS → lower discount rate → more capital required.

KEY FORMULAS — Quick Reference

Formula	Application
$E(Re) = D/P + g$	GGM equity return (long-run, g = nominal GDP growth)
$E(Re) = (D/P - \%\Delta S) + \%\Delta E + \%\Delta(P/E)$	Grinold-Kroner
$RP(i) = \omega \cdot RP(integrated) + (1-\omega) \cdot RP(segmented)$	Singer-Terhaar
$RP(integrated) = \rho(i,GM) \times \sigma(i) \times SR(GM)$	Fully integrated risk premium
$RP(segmented) = \sigma(i) \times SR(i)$	Fully segmented risk premium
$U = E(R) - 0.005\lambda\sigma^2$	MVO utility function
$r(Taylor) = r(n) + \pi(e) + 0.5(GDP\ gap) + 0.5(\pi\ gap)$	Taylor Rule
$E(r,RE) = NOI/P + g - \%\Delta(Cap\ Rate)$	Real estate return (short-run)
$MCTR = \beta(i) \times \sigma(P)$	Marginal contribution to total risk
$ACTR = w(i) \times MCTR(i)$	Absolute contribution to total risk
$Surplus = MVa - PVL$	Liability-relative
$var(true) = [(1+\lambda)/(1-\lambda)] \times var(observed)$	De-smoothing
$r(at) = r(pt) \times (1-t)$	After-tax return on interest
$\sigma(at) = \sigma(pt) \times (1-t)$	After-tax volatility

Generated from Mark Meldrum 2026 L3 Notes + Richard's Mock Exam Mistakes